

2(a)	peptidoglycan / murein ; ribosomes ; (permanent / large) vacuole ;	3
2(b)	<i>any two from:</i> to prevent, lysis / bursting / AW or to prevent, (protoplast) shrinking / shrivelling / AW ; I damaged <i>ref. to</i> no net movement / AW, of water (occurs) ; (otherwise movement of water) by <u>osmosis</u> / down a water potential gradient / AW ;	2

Question	Answer	Marks
2(c)(i)	<i>max 2 if stated as a protein or includes protein features</i> <i>any three from:</i> <i>cellulose molecule max 2</i> (β-glucose(s) joined by) glycosidic / glucosidic, bonds / linkages ; (monomers / β-glucose(s)) joined by (β) 1,4 (glycosidic) bonds ; R α-1,4 bonds R (β)1,4 <u>and</u> 1,6 bonds adjacent, monomers / β-glucoses, rotated through 180° ; AW straight / linear, chain ; <i>cellulose microfibril max 2</i> parallel, molecules / chains, of cellulose ; hydrogen bonds, between molecules / form cross bridges / between OH groups ; <i>idea that</i> between adjacent cellulose molecules, beginnings and ends in different places / overlap between starts and ends / staggered starts and ends ;	3
2(c)(ii)	<i>any one from:</i> hemicellulose ; pectin ; lignin ; AVP ; e.g. expansin	1
2(d)	spongy mesophyll / palisade mesophyll / palisade / mesophyll, (cell) ;	1